DECISION SHEET

Standard(s)- (year and edition):	**Sub clause(s):**	**Sheet n°:**
IEC 60601-1:1988 Am1+Am2	57.6	**DSH-433**
Subject: Components between primary and secondary	**Key words:** Reinforced, insulation, bridging	**Confirmed by CTL at its 39th meeting, in Cologne**

Question:

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DOUBLE INSULATION and REINFORCED INSULATION are defined in subclasses 2.3.4 and 2.3.7. Relevant requirements are in sub-clause 20 (Dielectric strength) and sub-clause 57.10 (CREEPAGE DISTANCES and AIR CLEARANCES). Sometimes it is necessary to bridge the insulation by components (e.g. opto-couplers, capacitors). What are the requirements for these components?

Decision:

Capacitors according to IEC 60384-14 type Y1 can be used to bridge DOUBLE or REINFORCED INSULATION.

Step by step a list with references to standards for components with requirements concerning reliability shall be established.

Explanatory notes:

Most of the components used to bridge DOUBLE or REINFORCED INSULATION withstand relevant dielectric strength tests. As dielectric strength is a type test no evidence is given concerning reliability of the components.

Reliability of components is important to prevent break down of insulations during lifetime of EQUIPMENT. Just a few of the standards addressing requirements for components relate to reliability. Only components on the basis of such standards can be used to bridge DOUBLE or REINFORCED INSULATION.